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Enantioselective addition of diethylzinc to aldehydes catalyzed by aziridine carbinols

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ABSTRACT

A series of enantiopure aziridine carbinols were synthesized from commercially available starting materials (L-serine and piperonylaldehyde), and characterized by ¹H NMR, ¹³C NMR, HRMS, and IR. Their enantioselective induction behaviors were evaluated as chiral ligands in the asymmetric addition of diethylzinc and phenylethynylzinc to aryl and α,β -unsaturated aldehydes. Chiral aziridine carbinol **1h** was found to catalyze the asymmetric addition of diethylzinc to aldehydes at low catalyst loadings (as little as 1 mol %), and the conversion rates were close to quantitative (up to 97% yield) with good to excellent enantiomeric excesses (up to 98% ee). A possible catalytic reaction mechanism is proposed.

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1. Introduction

The enantioselective addition of organozinc reagents to aldehydes is known as one of the most valuable ways to obtain enantiomerically pure secondary alcohols via C–C bond formation. It has received widespread attention because these alcohols are not only components of many naturally occurring and biologically active compounds, but also valuable intermediates for the syntheses of other functionalities, such as halides, amides, esters, and ethers.¹ Since the pioneering work of Oguni and Omi, where chiral β -amino alcohols were used to catalyze the addition of diethylzinc to benzaldehyde with 49% ee, and moreover, the seminal contributions to this field by Noyori et al. in the late 20th century, significant efforts have been devoted to the development of more selective and efficient catalytic systems for this synthetically useful transformation. Later, the asymmetric addition of diethylzinc to aldehydes has become a benchmark standard for testing the effectiveness of new chiral catalysts.²

Despite the excellent results that have been obtained using diverse ligand structures (amino alcohols,³ diols,⁴ diamines⁵), a crucial factor influencing the chemical yield and stereoselectivity of asymmetric reactions is the use of an appropriate chiral catalyst. Hence, the design and development of novel backbones and efficient chiral ligands remain an enduring interest in the asymmetric synthesis.

Among chiral amine alcohols, small-ring nitrogen heterocyclic alcohols constitute as a very interesting class of compounds when applied as chiral ligands, especially in asymmetric additions of organozinc species to carbonyl compounds.⁶ In recent years, our group has paid continuing attention to the synthesis and application of chiral small-ring nitrogen heterocycle ligands containing a β -aminoalcohol scaffold in highly enantioselective addition reactions.⁷ Recently, the quantitative relationship between the enantioselectivity with the conformations of catalysts in the addition of diethylzinc to benzaldehydes were reported by us; our conclusions offered a theoretical basis for chiral catalyst design, evaluation, and prediction.⁸ We also previously reported that various *N*-substituted moiety aziridino diphenylmethanols were effective ligands for copper(II)-catalyzed asymmetric Henry reactions, and novel recyclable chiral dendritic ligands in asymmetric diethylzinc additions to aldehydes with excellent enantioselectivities.⁹ In continuation of our current interests in the field of asymmetric synthesis, especially in the application of aziridine-based chiral catalysts, and taking into account the fact that aziridines strongly coordinate to organozinc species, we herein report the synthesis of various *N*-piperonyl moiety aziridine carbinols **1** and their catalytic activity evaluation toward the addition of diethylzinc to aromatic aldehydes.

2. Results and discussion

2.1. Synthesis of the aziridine carbinols

Enantiopure *N*-piperonyl aziridine carbinols (Fig. 1) were obtained in good overall yields via Grignard reaction using *N*-piperonylaziridine-2-carboxylic esters, according to the

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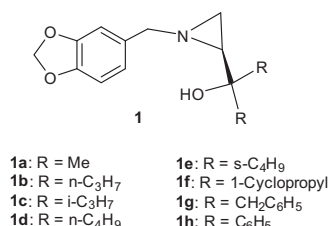


Figure 1. N-Piperonyl moiety chiral ligands.

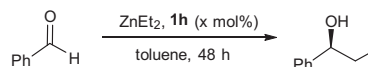
literature;⁹ **1a** (80%), **1b** (84%), **1c** (73%), **1d** (81%), **1e** (77%), **1f** (92%), **1g** (89%), **1h** (92%).

2.2. Screening of the ligands

The catalytic activity and the enantioselectivity of the aziridine carbinol ligands **1a–1h** (5 mol %) were studied using the reaction of diethylzinc and benzaldehyde in toluene at 0 °C as a model reaction (Table 1). The study on the substituent effect of ligands identified that the introduction of two aromatic groups onto the ligand led to a dramatic increase in the yield and enantioselectivity (entries 7 and 8, in Table 1). In particular, the use of ligand **1h** bearing two phenyl groups resulted in the formation of an (S)-secondary alcohol in 84% yield with 92% ee (entry 8). On the other hand, ligands **1a–1f** bearing alkyl groups showed moderate or poor yields and catalytic activities (entries 1–6). Therefore, ligand **1h** was determined as the most effective ligand, and further studies were conducted. The amount of chiral ligand and the reaction temperature were screened for further optimization of the reaction conditions using ligand **1h** (Table 2). When the amount of ligand **1h** was increased from 1% to 10% in toluene at 0 °C (Table 2, entries 1–5), there was no obvious change in the yield or the enantioselectivity of this reaction, which indicated that the amount of chiral ligand had no significant influence on the catalytic reaction in this range. The reaction temperature was then optimized under the lowest amount of ligand **1h** (1 mol %). The results showed that the reaction temperature played an important role in the yield and ee value; a higher or lower reaction temperature had a negative influence on the result (Table 2, entries 6–8). When the reaction temperature increased from 0 °C to room temperature (Table 2, entry 6), the ee value decreased from 94% to 49%. When the temperature was –10 °C or –20 °C (Table 2, entries 7

Table 2

Optimization of the asymmetric addition of diethylzinc to benzaldehyde catalyzed by **1h**^a



Entry	1h (x mol %)	Temp (°C)	Time (h)	Yield ^b (%)	ee ^c (%)	Config. ^d
1	1	0	48	95	94	(S)
2	3	0	48	82	94	(S)
3	5	0	48	84	92	(S)
4	7	0	48	85	90	(S)
5	10	0	48	85	94	(S)
6	1	rt	48	85	49	(S)
7	1	–10	48	55	85	(S)
8	1	–20	48	31	65	(S)

^a The reaction was carried out in toluene with 3.0 equiv of diethylzinc (1.0 M solution in hexane) to benzaldehyde.

^b Isolated yields by TLC.

^c Determined by HPLC using a chiral OD-H column.

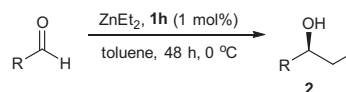
^d Absolute configuration was assigned by known elution order from chiral column according to the literature.

and 8), the yields and enantioselectivities all decreased significantly (Table 2, entry 8). Overall, the optimized reaction conditions were found to be as little as 1 mol % of chiral ligand **1h**, in toluene at 0 °C for 48 h.

In order to examine the substrate scope of this transformation by ligand **1h** under the optimum reaction conditions mentioned above, the reactions were carried out using various aromatic aldehydes, and the results are shown in Table 3. Since the yields were determined after isolation of the products, small deviations might be caused by preparative manipulations. The results suggested that the substituents on the benzene ring of the benzaldehyde derivatives (Table 3, entries 1–10) did not significantly influence the ee values of the products, and the enantiomeric purities of the products were between 90% and 98%. The highest ee (98% ee) was

Table 3

Enantioselective addition of diethylzinc to various aldehydes in the presence of ligand **1h**^a



Entry	RCHO	Prod.	Yield ^b (%)	ee ^c (%)	Config. ^d
1	PhCHO	2a	95	94	(S)
2	<i>p</i> -MeC ₆ H ₄ CHO	2b	89	97	(S)
3	<i>o</i> -MeOC ₆ H ₄ CHO	2c	66	96	(S)
4	<i>m</i> -MeOC ₆ H ₄ CHO	2d	61	90	(S)
5	<i>p</i> -MeOC ₆ H ₄ CHO	2e	85	90	(S)
6	<i>o</i> -ClC ₆ H ₄ CHO	2f	90	92	(S)
7	<i>p</i> -ClC ₆ H ₄ CHO	2g	95	98	(S)
8	1-Naphthaldehyde	2h	81	94	(S)
9	2-Naphthaldehyde	2i	81	96	(S)
10	3,4-OCH ₂ OC ₆ H ₃ CHO	2j	97	96	(S)
11	Ferrocenealdehyde	2k	95	93	(S)
12	<i>trans</i> -Cinnamaldehyde	2l	78	74	(S)
13 ^e	PhCHO	2m	80	5	(S)

^a The reaction was carried out under the optimized conditions.

^b Isolated yields by TLC.

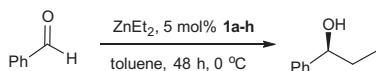
^c Determined by HPLC using a Chiralpak OD-H or OB-H column. The product chromatograms were compared against a known racemic mixture.

^d Absolute configuration was assigned by known elution order from chiral column according to the literature.

^e The addition of phenylacetylene (3.0 equiv) to benzaldehyde was examined in the presence of diethylzinc (3.0 equiv); the other reaction conditions remained the same.

Table 1

Screening of ligands **1a–h** for the enantioselective addition of diethylzinc to benzaldehyde^a



Entry	Ligand	Yield ^b (%)	ee ^c (%)	Config. ^d
1	1a	33	40	(S)
2	1b	52	76	(S)
3	1c	74	79	(S)
4	1d	52	45	(S)
5	1e	70	56	(S)
6	1f	76	87	(S)
7	1g	81	87	(S)
8	1h	84	92	(S)

^a The reaction was carried out in toluene with 5 mol % ligand, 3.0 equiv of diethylzinc (1.0 M solution in hexane) to benzaldehyde.

^b Isolated yields by TLC.

^c Determined by HPLC using a chiral OD-H column.

^d Absolute configuration was assigned by known elution order from chiral column according to the literature.

obtained when *p*-chlorobenzaldehyde was used as the substrate (Table 3, entry 7). In addition, good yields were observed with various aromatic aldehydes (Table 3, entries 1–11), except in the case of 2-methoxybenzaldehyde and 3-methoxybenzaldehyde (Table 3, entries 3 and 4), in which the yields dropped to 66% and 61%, possibly due to the electron-donating effect and steric hindrance of the methoxyl group. Moreover, ferrocenecarboxaldehyde also reacted smoothly with diethylzinc under these conditions with excellent yield and ee values (Table 3, entry 11). The α,β -unsaturated aldehyde *trans*-cinnamaldehyde was also investigated under the same reaction conditions; unfortunately the enantioselectivity was lower compared with that obtained from aromatic aldehydes (Table 3, entry 12). Ligand **1h** was also tested in the asymmetric addition of phenylethynylzinc to benzaldehyde, leading to propargylic alcohol in good chemical yield with poor enantioselectivity (Table 3, entry 13).

Based on the experimental results, theoretical analysis and the related mechanism suggested by us and Corey,^{9,10} a plausible mechanism for the enantioselective addition of diethylzinc to arylaldehydes catalyzed by ligand **1h** is proposed (Fig. 2). Ligand **1h** forms a five-membered cyclic adduct (zinc monoalkoxide **A**) with an ethylzinc moiety. This adduct is stable and rigid enough to accept a diethylzinc molecule via the oxygen atom to form the metalocyclic catalyst **B**.

After coordination to benzaldehyde, an ethyl group of the diethylzinc molecule can attack the *Si*-face and *Re*-face of the carbonyl group of PhCHO and lead to transition states **C** and **D**, respectively. Transition state **D** is disfavored because of steric hindrance (designated by arrows in Fig. 2). Consequently, transition state **C**, which is the favored structure, leads to the major enantiomer (*S*)-1-arylpropanol. On the other hand, the minor enantiomer (*R*)-1-arylpropanol might be formed by the unfavored transition state **D**. In addition, the piperonylmethyl group was essential for obtaining higher enantioselectivity, maybe due to its hindrance and electron pair effect, which is in agreement with our previous reports.^{9a}

3. Conclusion

In conclusion, we have reported the application of enantiopure aziridine carbinols in the catalytic asymmetric enantioselective addition of diethylzinc to aldehydes. The enantiopure ligands, which were easily synthesized through a five-step procedure with commercially available starting materials, were successfully used in the enantioselective addition of diethylzinc to aldehydes and afforded chiral secondary alcohols in good reactivity and enantioselectivity. It should also be noted that with 1 mol % ligand **1h**, the less polar aromatic solvent toluene was the most effective for asymmetric induction. Further applications of these ligands to other asymmetric reactions are currently underway in our laboratory.

4. Experimental

4.1. General

Melting points were determined using YRT-3 melting point apparatus and are uncorrected. Optical rotations were measured with Perkin Elmer, model 341 Polarimeter at 20 °C in CH₂Cl₂. The enantiomeric purity was determined by HPLC using a chiral column with hexane/propan-2-ol (ratio as indicated) as the eluent. The chromatographic system consisted of a Varian prostar model 210 intelligent HPLC or Agilent 1260 Infinity HPLC with UV–vis detector (254 nm). The injection loop had a 20 μ L capacity. The column used was a Chiralcel OD-H or OB-H (250 $\times$ 4.6 mm) from Daicel Chemical Ind., Ltd (Japan). The column was operated at ambient temperature. TLC was performed on dry silica gel plates developed with hexane/ethyl acetate. NMR spectra (¹H and ¹³C) were performed on a Bruker DPX 400 (400 MHz) spectrometer using solutions in CDCl₃ (referenced internally to Me₄Si). *J* values are given in Hz. IR spectra were determined on a Perkin Elmer FTIR 1750 spectrophotometer. Mass spectra were obtained using a Bruker esquire-3000 instrument with an electrospray ionization source (ESIMS). All the ESIMS spectra were performed using

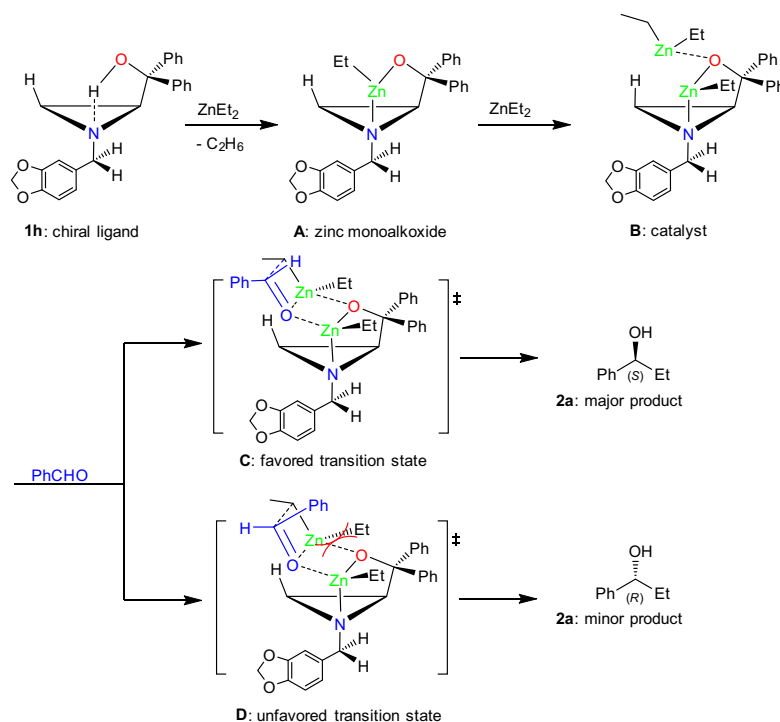


Figure 2. A plausible transition state for the addition of diethylzinc to benzaldehyde.

MeOH as the solvent. Compounds **1a–1h** were synthesized according to the literature procedure.⁹

4.1.1. (2S)-1-Piperonylmethylaziridin-2-yl(dimethyl)methanol **1a**

Light-yellow oil, yield 80%, $[\alpha]_D^{20} = +4.5$ (c 0.594, CHCl₃); ¹H NMR (400 MHz, CDCl₃) δ (ppm) 6.87 (s, 1H), 6.81–6.71 (m, 2H), 5.95 (s, 2H), 3.24, 3.54 (d, *J* = 12.8 Hz, 2H), 2.41 (d, *J* = 16.5 Hz, 1H), 1.88 (d, *J* = 3.6 Hz, 1H), 1.57 (dd, *J* = 6.4, 3.6 Hz, 1H), 1.37 (d, *J* = 6.4 Hz, 1H), 1.11 (d, *J* = 16.9 Hz, 6H); ¹³C NMR (100 MHz, CDCl₃) δ (ppm) 147.38, 146.50, 132.73, 121.32, 108.77, 107.82, 100.71, 66.64, 63.58, 46.65, 30.00, 28.93, 25.58; HRMS *m/z* (ESI) calcd for C₁₃H₁₇NO₃ [M⁺]: 235.1208, found: 235.1239; IR (KBr pellet) ν (cm^{−1}) 3430, 2974, 2892, 1856, 1612, 1495, 1446, 1369, 1247, 1091, 1037, 934, 864, 808, 569.

4.1.2. (2S)-1-Piperonylmethylaziridin-2-yl(di-*n*-propyl)methanol **1b**

Light-yellow oil, yield 84%, $[\alpha]_D^{20} = -12.7$ (c 1.05, CH₂Cl₂); ¹H NMR (CDCl₃, 400 MHz) δ (ppm) 6.86 (s, 1H), 6.78–6.72 (m, 2H), 5.94 (s, 2H), 3.38 (dd, *J* = 32.8, 12.8 Hz, 2H), 2.41 (s, 1H), 1.87 (d, *J* = 3.6 Hz, 1H), 1.58 (dd, *J* = 6.4, 3.6 Hz, 1H), 1.42–1.22 (m, 9H), 0.89 (t, *J* = 6.8 Hz, 3H), 0.81 (t, *J* = 6.8 Hz, 3H); ¹³C NMR (CDCl₃, 100 MHz) δ (ppm) 147.37, 146.51, 132.61, 121.32, 108.79, 107.79, 100.69, 69.85, 63.30, 44.32, 42.50, 39.44, 29.67, 16.59, 16.41, 14.57, 14.46; HRMS *m/z* (ESI) calcd for C₁₇H₂₅NO₃ [M+H⁺]: 292.1868, found: 292.1906; IR (KBr pellet) ν (cm^{−1}) 3449, 2955, 2878, 1494, 1446, 1369, 1246, 1187, 1039, 981, 864, 807.

4.1.3. (2S)-1-Piperonylmethylaziridin-2-yl(di-*i*-propyl)methanol **1c**

Light-yellow oil, yield 73%, $[\alpha]_D^{20} = -117.1$ (c 0.564, CHCl₃); ¹H NMR (CDCl₃, 400 MHz) δ (ppm) 6.86 (s, 1H), 6.80–6.72 (m, 2H), 5.95 (s, 2H), 4.12 (d, *J* = 13.1 Hz, 1H), 2.74 (d, *J* = 13.1 Hz, 1H), 2.50 (br, 1H), 2.08–1.95 (m, 2H), 1.90 (d, *J* = 3.2 Hz, 1H), 1.60 (dd, *J* = 6.0, 3.8 Hz, 1H), 1.32 (d, *J* = 6.4 Hz, 1H), 1.02–0.90 (m, 12H); ¹³C NMR (CDCl₃, 100 MHz) δ (ppm) 147.43, 146.50, 121.10, 108.59, 107.84, 100.72, 72.62, 62.10, 40.53, 34.69, 33.11, 29.35, 17.48, 17.28, 16.98; HRMS *m/z* (ESI) calcd for C₁₇H₂₅NO₃ [M+H⁺]: 292.1868, found: 292.1900; IR (KBr pellet) ν (cm^{−1}) 3463, 2971, 2882, 1612, 1494, 1447, 1245, 1040, 930, 808.

4.1.4. (2S)-1-Piperonylmethylaziridin-2-yl(di-*n*-butyl)methanol **1d**

Light-yellow oil, yield 81%; $[\alpha]_D^{20} = -20.2$ (c 1.05, CH₂Cl₂); ¹H NMR (CDCl₃, 400 MHz) δ (ppm) 6.87 (s, 1H), 6.79–6.71 (m, 2H), 5.94 (s, 2H), 3.46 (d, *J* = 12.8 Hz, 1H), 3.31 (d, *J* = 12.8 Hz, 1H), 2.43 (s, 1H), 1.91 (d, *J* = 3.8 Hz, 1H), 1.60 (dd, *J* = 6.4, 3.8 Hz, 1H), 1.37–1.11 (m, 13H), 0.89 (t, *J* = 6.8 Hz, 3H), 0.83 (t, *J* = 6.8 Hz, 3H); ¹³C NMR (CDCl₃, 100 MHz) δ (ppm) 147.38, 146.54, 132.52, 121.40, 108.84, 107.78, 100.70, 69.85, 63.39, 44.48, 39.71, 36.75, 29.76, 25.45, 25.28, 23.18, 23.07, 13.87, 13.78; HRMS *m/z* (ESI) calcd for C₁₉H₂₉NO₃ [M+H⁺]: 320.2181, found: 320.2224; IR (KBr pellet) ν (cm^{−1}) 3465, 2942, 2866, 1495, 1447, 1370, 1329, 1093, 1040, 932, 803.

4.1.5. (2S)-1-Piperonylmethylaziridin-2-yl(di-*i*-butyl)methanol **1e**

Yellow oil, yield 77%, $[\alpha]_D^{20} = -61.0$ (c 1.05, CH₂Cl₂); ¹H NMR (CDCl₃, 400 MHz) δ (ppm) 6.81 (s, 1H), 6.77–6.67 (m, 2H), 5.90 (s, 2H), 3.98 (ddd, *J* = 21.0, 13.1, 7.4 Hz, 1H), 2.75 (ddd, *J* = 21.0, 13.1, 7.4 Hz, 1H), 1.84 (m, 1H), 1.75–1.49 (m, 5H), 1.28–0.75 (m, 16H); ¹³C NMR (CDCl₃, 100 MHz) δ (ppm) 147.42, 146.48, 132.38, 121.09, 108.59, 107.82, 100.70, 73.34, 62.45, 62.16, 42.37, 42.10, 41.36, 40.78, 29.33, 23.55, 23.15, 13.34, 12.74; HRMS *m/z* (ESI) calcd for C₁₉H₂₉NO₃ [M+H⁺]: 320.2181, found: 320.2224; IR (KBr

pellet) ν (cm^{−1}) 3469, 2967, 2881, 1494, 1449, 1371, 1322, 1246, 1186, 1099, 1040, 937, 864, 806.

4.1.6. (2S)-1-Piperonylmethylaziridin-2-yl(dicyclopropyl)methanol **1f**

Light-yellow oil, yield 92%, $[\alpha]_D^{20} = -27.2$ (c 1.05, CH₂Cl₂); ¹H NMR (CDCl₃, 400 MHz) δ (ppm) 6.85 (s, 1H), 6.78–6.72 (m, 2H), 5.94 (s, 2H), 3.40 (dd, *J* = 32.8, 12.8 Hz, 2H), 2.22 (s, 1H), 1.88 (d, *J* = 3.6 Hz, 1H), 1.71 (dd, *J* = 6.4, 3.6 Hz, 1H), 1.34 (d, *J* = 6.5 Hz, 1H), 0.84 (tt, *J* = 8.5, 5.6 Hz, 1H), 0.72 (tt, *J* = 8.5, 5.6 Hz, 1H), 0.51–0.07 (m, 8H); ¹³C NMR (CDCl₃, 100 MHz) δ (ppm) 147.56, 146.70, 132.68, 121.67, 109.14, 107.98, 100.88, 66.96, 63.40, 45.42, 29.11, 18.57, 17.50; HRMS *m/z* (ESI) calcd for C₁₇H₂₁NO₃ [M+H⁺]: 288.1555, found: 288.1597; IR (KBr pellet) ν (cm^{−1}) 3433, 3080, 3006, 2890, 1495, 1442, 1372, 1312, 1246, 1196, 1036, 927, 862, 809.

4.1.7. (2S)-1-Piperonylmethylaziridin-2-yl(dibenzyl)methanol **1g**

White solid, yield 89%, mp = 84.6–85.8 °C; $[\alpha]_D^{20} = -24.3$ (c 1.05, CH₂Cl₂); ¹H NMR (400 MHz, CDCl₃) δ (ppm) 7.37–7.18 (m, 10H), 6.69 (d, *J* = 7.9 Hz, 1H), 6.59 (s, 1H), 6.52 (d, *J* = 7.9 Hz, 1H), 5.91 (s, 2H), 3.21 (s, 1H), 3.05 (d, *J* = 12.9 Hz, 1H), 2.92 (d, *J* = 13.5 Hz, 1H), 2.85 (s, 2H), 2.78 (d, *J* = 13.5 Hz, 1H), 2.58 (d, *J* = 12.8 Hz, 1H), 1.64 (dd, *J* = 5.9, 3.7 Hz, 1H), 1.46 (d, *J* = 3.3 Hz, 1H), 1.15 (d, *J* = 6.4 Hz, 1H); ¹³C NMR (CDCl₃, 100 MHz) δ (ppm) 147.50, 146.66, 137.63, 137.25, 130.73, 130.63, 128.07, 127.82, 126.32, 126.22, 121.31, 108.80, 107.90, 100.86, 70.79, 61.91, 47.16, 45.36, 43.48, 29.43; HRMS *m/z* (ESI) calcd for C₂₅H₂₅NO₃ [M+H⁺]: 388.1834, found 388.5026.

4.1.8. (2S)-1-Piperonylmethylaziridin-2-yl(diphenyl)methanol **1h**

White solid, yield 92%, mp = 91.4–92.8 °C; $[\alpha]_D^{20} = -2.0$ (c 1.05, CH₂Cl₂); ¹H NMR (400 MHz, CDCl₃) δ (ppm) 7.48–7.39 (m, 2H), 7.31–7.28 (m, 4H), 7.27–7.17 (m, 4H), 6.70–6.61 (m, 3H), 5.93 (s, 2H), 3.77 (s, 1H), 3.48 (dd, *J* = 32.5, 12.9 Hz, 2H), 2.50 (dd, *J* = 6.3, 3.5 Hz, 1H), 2.02 (d, *J* = 3.5 Hz, 1H), 1.61–1.50 (m, 2H); ¹³C NMR (CDCl₃, 100 MHz) δ (ppm) 147.52, 147.46, 146.75, 144.88, 132.22, 128.04, 127.93, 127.05, 126.68, 126.35, 126.22, 121.45, 108.84, 107.96, 100.89, 74.06, 62.89, 45.92, 30.58; HRMS *m/z* (ESI) calcd for C₂₃H₂₁NO₃ [M+H⁺]: 360.1555, found 360.1591; IR (KBr pellet) ν (cm^{−1}) 3350, 3024, 2980, 2888, 1491, 1444, 1368, 1242, 1193, 1037, 989, 931, 778, 760, 696.

4.2. General procedure for the enantioselective diethylzinc addition to arylaldehydes

To a stirring mixture of ligand **1h** (0.005 mmol, 1 mol %) in toluene (5 mL), diethylzinc (1 M in *n*-hexane, 1.5 mL) was added under a nitrogen atmosphere. The mixture was cooled to 0 °C, and stirred for 0.5 h, followed by the addition of the corresponding freshly distilled aldehyde (0.5 mmol). The mixture was stirred for 48 h at the same temperature, and then quenched with saturated aqueous NH₄Cl (4 mL). The organic layer was separated and the aqueous phase was extracted with Et₂O (10 mL × 3). The organic layers were combined and dried over anhydrous MgSO₄ and evaporated under reduced pressure. The residue was purified by the preparative silica gel TLC plate (ethylacetate and petroleum ether, 1:6, v/v) to afford the corresponding chiral alcohol as a colorless oil or light-yellow solid. The enantiomeric excess was determined by HPLC analysis using a Chiralcel OD-H or OB-H column. Conditions of HPLC analysis: Chiralcel OD-H and Chiralcel OB-H 4.6 mm × 250 mm; detection, 254 nm light. The absolute configurations of the products were assigned by comparison to the literature data. In all cases, the product chromatograms were compared against a known racemic mixture.

4.2.1. (S)-1-Phenyl-1-propanol 2a

95% isolated yield, 94% ee by HPLC analysis (Chiralcel OD-H column, eluent: hexane/*i*-PrOH, 100:2; flow rate: 1.0 mL/min). Retention times: 16.0 min [major (*S*)-enantiomer] and 13.2 min [minor (*R*)-enantiomer].

4.2.2. (S)-1-*p*-Tolyl-1-propanol 2b

89% isolated yield, 97% ee by HPLC analysis (Chiralcel OB-H column, eluent: hexane/*i*-PrOH, 95:5; flow rate: 0.5 mL/min). Retention times: 13.1 min [major (*S*)-enantiomer] and 18.0 min [minor (*R*)-enantiomer].

4.2.3. (S)-1-(2-Methoxyphenyl)-1-propanol 2c

66% isolated yield, 96% ee by HPLC analysis (Chiralcel OD-H column, eluent: hexane/*i*-PrOH, 100:1; flow rate: 1.0 mL/min). Retention times: 22.8 min [major (*S*)-enantiomer] and 25.7 min [minor (*R*)-enantiomer].

4.2.4. (S)-1-(3-Methoxyphenyl)-1-propanol 2d

61% isolated yield, 90% ee by HPLC analysis (Chiralcel OD-H column, eluent: hexane/*i*-PrOH, 100:2; flow rate: 1.0 mL/min). Retention times: 33.9 min [major (*S*)-enantiomer] and 30.1 min [minor (*R*)-enantiomer].

4.2.5. (S)-1-(4-Methoxyphenyl)-1-propanol 2e

85% isolated yield, 90% ee by HPLC analysis (Chiralcel OD-H column, eluent: hexane/*i*-PrOH, 100:1; flow rate: 1.0 mL/min). Retention times: 39.7 min [major (*S*)-enantiomer] and 34.9 min [minor (*R*)-enantiomer].

4.2.6. (S)-1-(2-Chlorophenyl)-1-propanol 2f

90% isolated yield, 92% ee by HPLC analysis (Chiralcel OB-H column, eluent: hexane/*i*-PrOH, 95:5; flow rate: 0.5 mL/min). Retention times: 10.2 min [major (*S*)-enantiomer] and 13.8 min [minor (*R*)-enantiomer].

4.2.7. (S)-1-(4-Chlorophenyl)-1-propanol 2g

95% isolated yield, 98% ee by HPLC analysis (Chiralcel OD-H column, eluent: hexane/*i*-PrOH, 100:1; flow rate: 0.5 mL/min). Retention times: 35.9 min [major (*S*)-enantiomer] and 38.2 min [minor (*R*)-enantiomer].

4.2.8. (S)-1-(1'-Naphthyl)-1-propanol 2h

81% isolated yield, 94% ee by HPLC analysis (Chiralcel OD-H column, eluent: hexane/*i*-PrOH, 95:5; flow rate: 1.0 mL/min). Retention times: 14.4 min [major (*S*)-enantiomer] and 31.3 min [minor (*R*)-enantiomer].

4.2.9. (S)-1-(2'-Naphthyl)-1-propanol 2i

81% isolated yield, 96% ee by HPLC analysis (Chiralcel OD-H column, eluent: hexane/*i*-PrOH, 90:10; flow rate: 0.5 mL/min). Retention times: 21.5 min [major (*S*)-enantiomer] and 24.1 min [minor (*R*)-enantiomer].

4.2.10. (S)-1-Piperonyl-1-propanol 2j

97% isolated yield, 96% ee by HPLC analysis (Chiralcel OD-H column, eluent: hexane/*i*-PrOH, 98:2; flow rate: 0.5 mL/min). Retention times: 36.9 min [major (*S*)-enantiomer] and 27.8 min [minor (*R*)-enantiomer].

4.2.11. (S)-1-Ferrocenyl-1-propanol 2k

95% isolated yield, 93% ee by HPLC analysis (Chiralcel OD-H column, eluent: hexane/*i*-PrOH, 100:2; flow rate: 1.0 mL/min). Retention times: 15.8 min [major (*S*)-enantiomer] and 14.9 min [minor (*R*)-enantiomer].

4.2.12. (1E,3S)-1-Phenylpent-1-en-3-ol 2l

78% isolated yield, 74% ee by HPLC analysis (Chiralcel OD-H column, eluent: hexane/*i*-PrOH, 90:10; flow rate: 1 mL/min). Retention times: 9.9 min [major (*S*)-enantiomer] and 7.1 min [minor (*R*)-enantiomer].

4.2.13. (S)-1,3-Diphenylprop-2-yn-1-ol 2m

80% isolated yield, 5% ee by HPLC analysis (Chiralcel OD-H column, eluent: hexane/*i*-PrOH, 90:10; flow rate: 1 mL/min). Retention times: 11.7 min [major (*S*)-enantiomer] and 19.8 min [minor (*R*)-enantiomer].

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